

CPU can be fine for tiny local jobs; GPU wins when batching or concurrency grows.

How to read it: one measured CodeLlama 7B Q4 chat request. The cards show active-device joules, latency, and peak VRAM; the trace locates Python client work, prefill, GPU inference, KV cache, sampling, and response parsing.

How to read it: FP16 is the 100% baseline for the exact same model; q8/q4 are compressed representations. Left: costs to minimize (VRAM, energy, latency). Right: benefits to preserve or improve (throughput, toy-score retention). Bars average only complete same-model FP16/q8/q4 triplets.

#	Heuristic
1	Benchmark the workload, not the model card.
2	If quality barely improves but joules rise, use the smaller model.
3	If VRAM is the deployment constraint, use q4/q8.
4	If the workload changes, re-benchmark; do not reuse one winner.
5	If concurrency grows, use GPU batching.
6	If the workload is tiny and local, CPU may be good enough.
7	Track tokens/joule, not just accuracy.